



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PHYSICS
PAPER 3 THEORY

6032/3
2 hours 30 minutes

JUNE 2025 SESSION

Additional materials:
Answer booklet
Electronic calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Write your answers on the separate answer booklet provided.

If you use more than one booklet, fasten the booklets/sheets together.

All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 10 printed pages.

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[Turn

Answer question 1 and any other 3 from the remaining questions.

- 1 (a) (i) Explain the term *derived units*.
- (ii) It is suggested that the period of oscillation of a floating cylinder in a liquid depends on the following quantities, its length, L , the density of the liquid, ρ , and the acceleration due to gravity, g .
- Derive an equation for the period of oscillation of the floating cylinder. [4]
- (b) (i) State the **two** conditions in which the equations of motion may be used.
- (ii) A car starting from rest travels for 1 minute with a uniform acceleration of 0.2 ms^{-2} . It then maintains a constant velocity for a further 3 minutes. The car is then brought to rest with a uniform deceleration of 0.4 ms^{-2} .
- Calculate the time taken for the journey. [5]
- (c) (i) State the conditions for a system of forces to be in *equilibrium*.
- (ii) Fig. 1.1 shows a uniform ladder 8 m long and of mass 40 kg resting in equilibrium with one end against a smooth vertical wall at an angle of 30° and the other on rough horizontal ground.

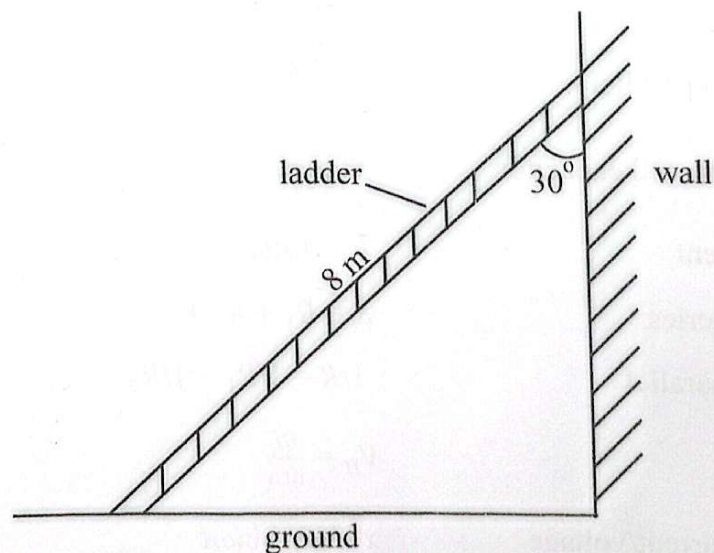


Fig. 1.1

1. Draw a free body diagram of the system showing all the forces.
 2. Calculate the force of friction between the ladder and the ground. [7]
- (d) (i) Distinguish between *electric potential energy* and *elastic potential energy*.

- (ii) A gas with a volume of 0.5 m^3 expands at a constant pressure of $2.0 \times 10^5 \text{ Pa}$ to double its volume.

Calculate the work done by the gas.

[3]

- (e) (i) Define *angular velocity*.

- (ii) Fig. 1.2 shows a mass of 100 g on the end of a length of string swinging in a horizontal circle of radius 30 cm . The string makes an angle of 45° with the vertical.

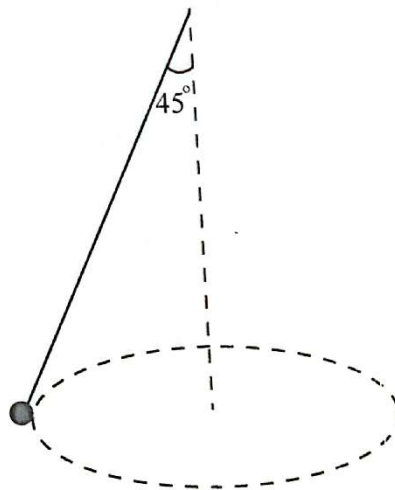


Fig. 1.2

Calculate the

1. tension in the string,
2. angular velocity of the mass,
3. period of the mass.

[6]

- 2 (a) (i) State the conditions necessary for the formation of stationary waves.
- (ii) Explain how these conditions are satisfied in the case of stationary waves in a pipe closed at one end.
- (iii) A double slit interference pattern was produced using slits separated by 0.45 mm when monochromatic light of wavelength 633 nm was illuminated. The pattern was projected on a screen 2.50 m from the slits.
- Calculate the fringe separation.

[7]

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[Turn over]



- (b) (i) Give **two** examples of waves that can be polarised.
 (ii) Explain what is meant by a *plane-polarised wave* and *unpolarised wave*. [4]

- (c) Sound waves vibrate with displacement, y , where

$$y = 0.20 \sin \left(100\pi t - \frac{10 - \pi x}{34} \right).$$

- (i) Determine the amplitude.
 (ii) Calculate the
 1. frequency of the sound waves,
 2. wavelength of the sound waves. [5]
- (d) (i) State **two** disadvantages and **two** advantages of using CT scanning for internal imaging.
 (ii) An X-ray beam transfers 20 kJ of energy through an area of 500 m² each second. The attenuation coefficient of bone is 300 m⁻¹.
 Calculate the intensity of the beam after passing through a 2.0 mm thickness of bone. [5]
- (e) (i) Explain the term *coherence of waves*.
 (ii) A diffraction grating has 500 lines per mm when it is illuminated normally by light of wavelength 550 nm.
 Calculate the maximum number of fringes that can be obtained on the screen. [4]

- 3 (a) (i) Define
 1. *charge*,
 2. *Coulomb*.
 (ii) Explain how electrostatics is used to extract dust particles from chimneys of factories. [5]



- (b) (i) State and explain *Kirchhoff's laws*.
- (ii) Two cells of electromotive force (e.m.f) 3.0 V and 1.5 V of negligible internal resistance, and two ammeters A_1 and A_2 , are connected in a network of resistors as shown in Fig. 3.1.

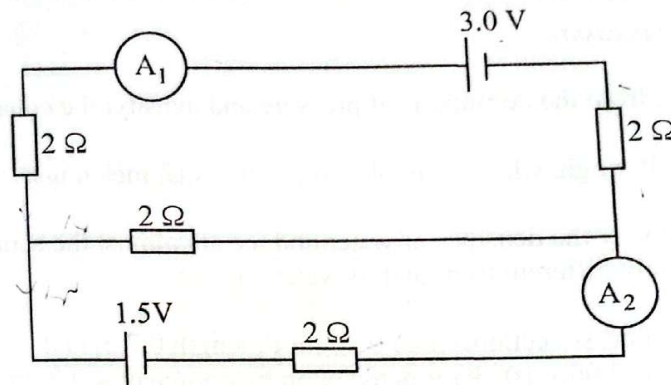


Fig. 3.1

Calculate the readings of ammeter A_1 and A_2 .

[8]

- (c) Fig 3.2 shows a wire of length 6.0 cm moving across a uniform magnetic field of flux density 1.2 mT at a velocity of 8.0 m/s.

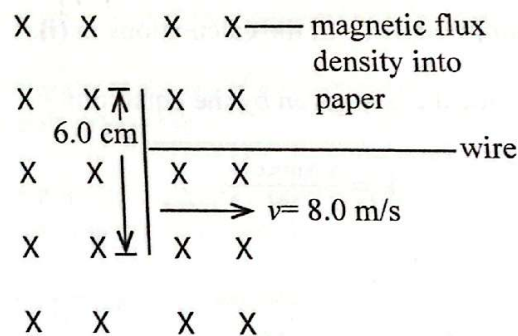


Fig. 3.2

- (i) Define *magnetic flux* and state its S.I unit.
- (ii) Calculate the magnitude of the induced e.m.f.
- (iii) State **three** ways in which the value of induced e.m.f can be increased.
- (d) (i) Explain the following properties of an ideal operational amplifier
1. infinite input impedance,
 2. infinite slew rate,
 3. infinite bandwidth.

[7]



- (ii) State the effect of negative feedback on the

1. gain,
2. bandwidth of an operational amplifier.

[5]

- 4 (a) (i) Define *pressure*.

- (ii) Derive, from the definitions of pressure and density, the equation:

$$P = \rho gh, \text{ where symbols have their usual meanings.}$$

- (iii) Explain why the densities of water and ice are almost the same but completely different from that of water vapour. [6]

- (b) A nylon string of cross sectional area 0.20 mm^2 , length 0.7 m and Young's Modulus $2.00 \times 10^9 \text{ Pa}$ is acted upon by a force of 4.0 N to increase its length.

- (i) Calculate the

1. percentage increase in length,
2. energy stored in the string when it is stretched.

- (ii) State the assumption made in the calculations in (i). [7]

- (c) The pressure, P , of an ideal gas is given by the equation:

$$P = \frac{1}{3} \frac{Nm\langle c^2 \rangle}{V}.$$

- (i) State the

1. significance of $\frac{1}{3}$ in the equation,
2. meaning of the product Nm .

- (ii) Using the given equation and the equation of state for an ideal gas, show that

$$\langle c^2 \rangle = \frac{3RT}{N_A m}, \text{ where symbols have their usual meanings.}$$

- (iii) Explain why the internal energy of an ideal gas is proportional to $\langle c^2 \rangle$. [7]

- (d) (i) State the *Bernoulli effect*.

- (ii) Use the Bernoulli effect to explain why air flow over the wings of an aircraft causes a lift force.



- (iii) State one condition under which the Bernoulli's principle is valid. [5]
- 5 (a) (i) *Electric charge can be quantised. Explain.*
- (ii) Explain how Millikan's experiment gives evidence of quantisation of charge.
- (iii) An electron is accelerated by a potential difference of 1000 V.
- Calculate the
1. work done,
 2. speed of the electron. [7]
- (b) (i) Explain the term *photoelectric effect*?
- (ii) Explain why photoelectrons are only emitted when incident radiation exceeds a certain value.
- (iii) A metal has a work function of 2.0 eV.
- Calculate the
1. maximum wavelength that causes photoemission,
 2. maximum velocity of the electrons emitted by a radiation of wavelength $0.1 \mu\text{m}$. [7]
- (c) (i) State and explain how the results from the alpha scattering experiment gives evidence for
1. the existence of a nucleus,
 2. the small size of the nucleus.
- (ii) Randon - $^{222}_{86}\text{Rn}$ decays to produce an isotope of Polonium $^{218}_{84}\text{Po}$.
- An equation of this decay is $^{222}_{86}\text{Rn} \rightarrow ^{218}_{84}\text{Po} + \text{X}$.
1. Identify particle X.
 2. Define the term *isotope*. [4]

- (d) (i) State any **five** advantages of using digital signals compared to analogue signals.
- (ii) Explain what is meant by a *geostationary satellite*.
- (iii) Explain the significance of a higher sampling frequency.